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DOP305-R

Building a CI/CD pipeline for blue/green deployments using AWS Fargate

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Building a CI/CD pipeline for blue/green deployments using AWS Fargate

Abstract:

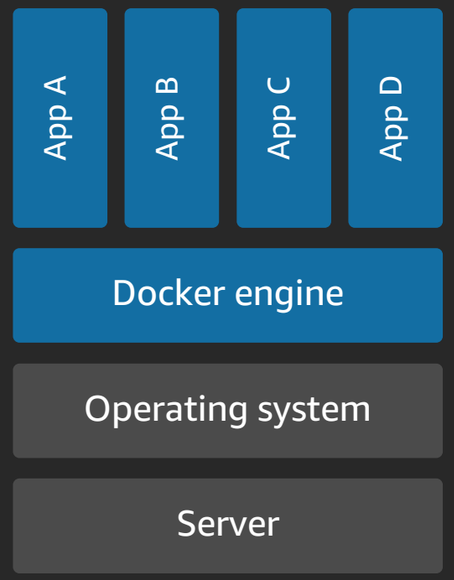
In this workshop you'll learn how to build a CI/CD pipeline (AWS CodePipeline) to develop a web-based application, containerize it, and deploy it on a serverless Amazon Elastic Container Service (ECS) cluster (AWS Fargate). You'll use the blue/green method to deploy it and you'll implement a switchover method using an AWS Lambda function. You will build this infrastructure using AWS CloudFormation, enabling you to reproduce the environment when needed. You'll also learn about monitoring and debugging capabilities using Amazon CloudWatch.

Agenda

- Containers and Docker
- Amazon ECS
- Fargate
- CI/CD pipeline with blue/green deployment method

Containers and Docker

A container is a standard unit of software that packages up code and all its dependencies, so the application runs quickly and reliably from one computing environment to another[1]



[1] https://www.docker.com/resources/what-container

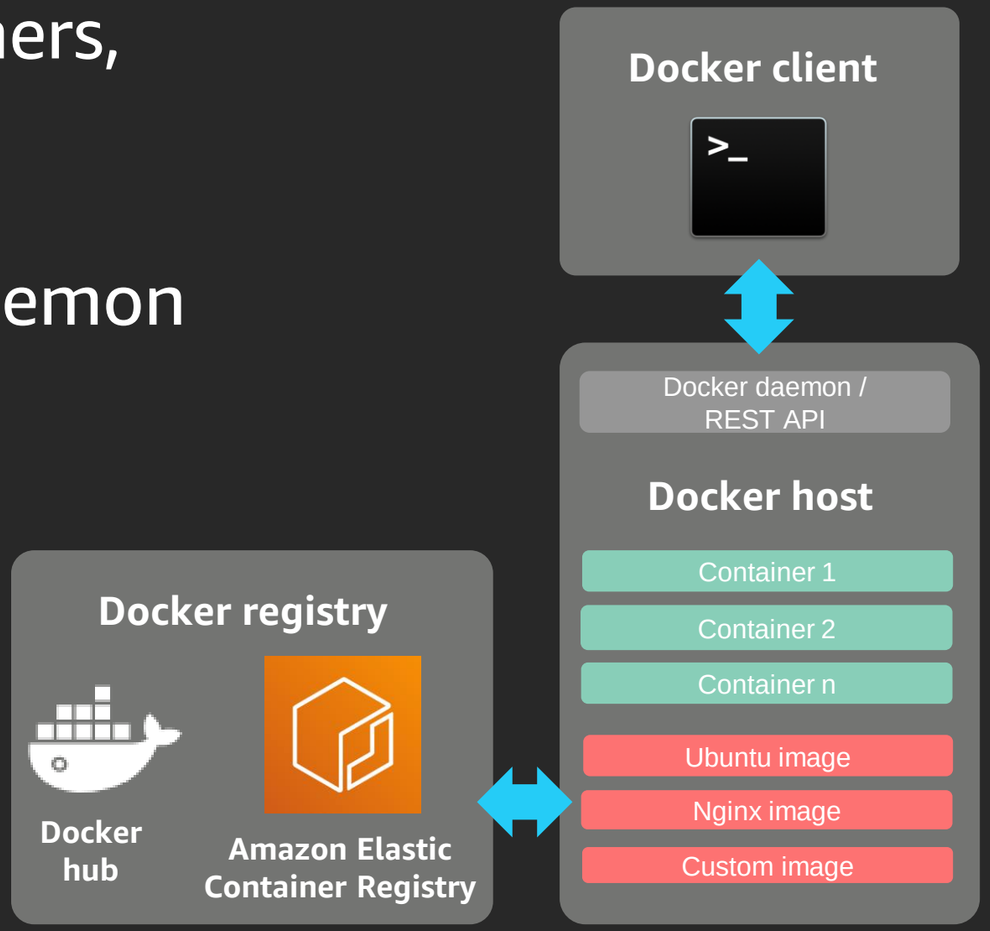
Docker architecture

Docker daemon manages images, containers, networks, and volumes

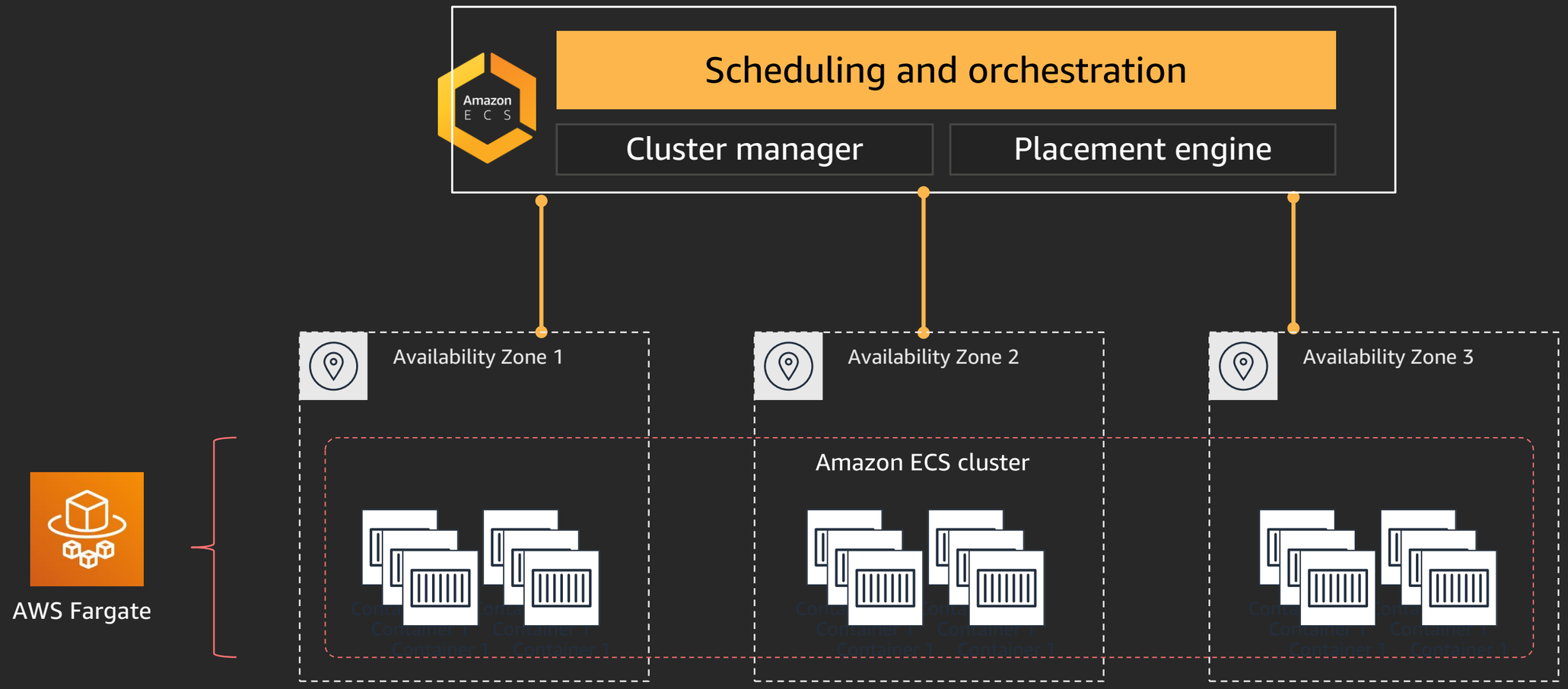
Docker CLI communicates with Docker daemon through a REST API

Sample commands:

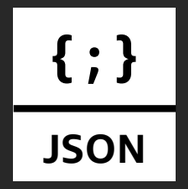
- `docker build` – Build image from Dockerfile
- `docker images` – List images on Docker host
- `docker run` – Run an image
- `docker ps` – List running containers
- `docker stop` – Stop a running container



Amazon ECS and Fargate



Amazon ECS constructs

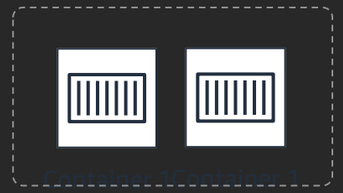


Task definition

- Template used by Amazon ECS to launch tasks
- Parallels to Docker run parameters
- Defines requirements:
  - CPU/Memory
  - Container image(s)
  - Logging
  - IAM role, etc.

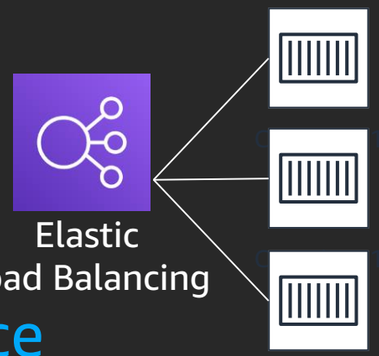
Cluster

- Resource grouping and isolation
- AWS Identity and Access Management (IAM) permissions boundary



Task

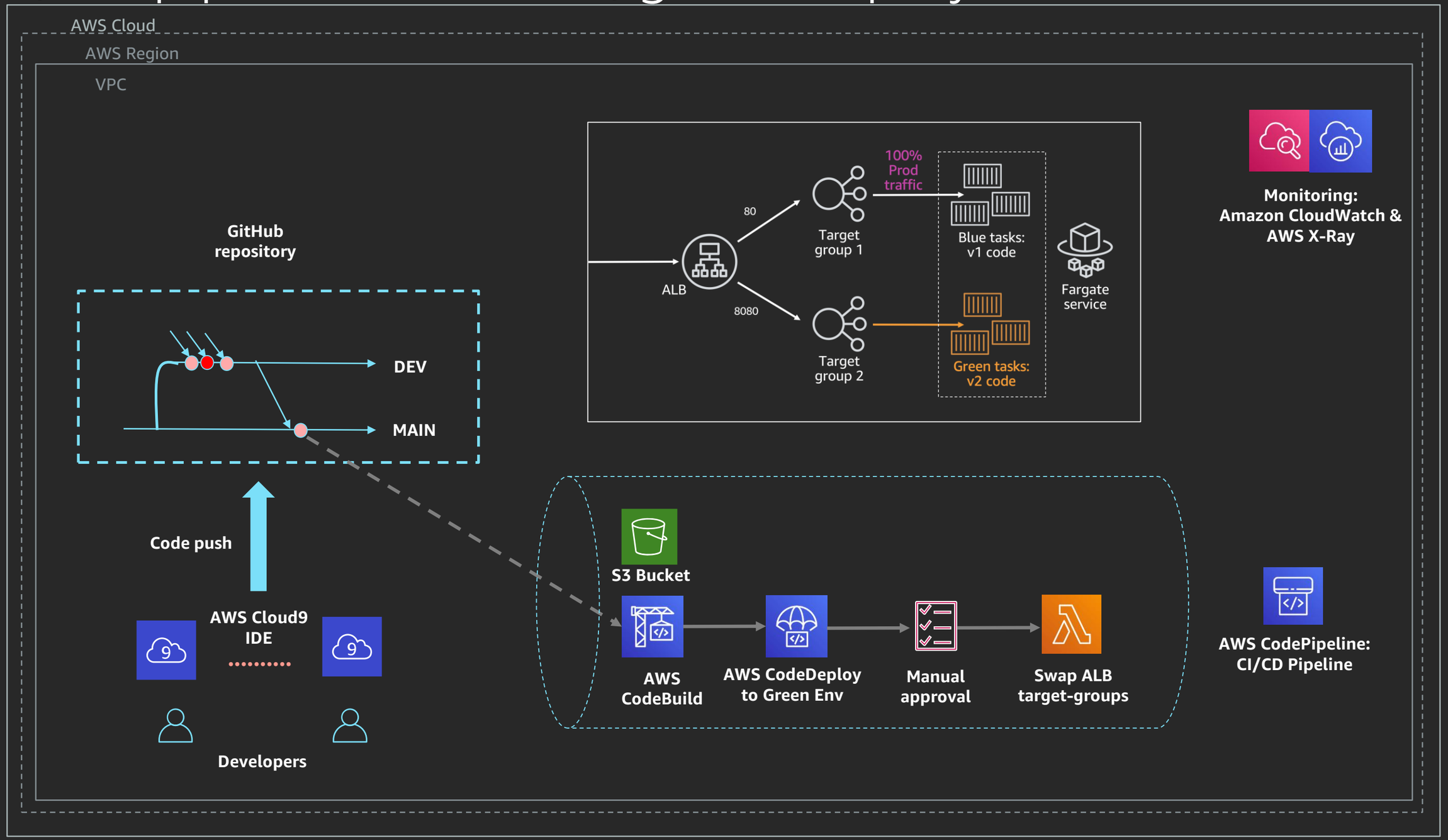
- Running instance of a task definition
- One or more containers



Service

- Maintains desired number of running tasks
- Replaces unhealthy tasks
- ELB integration

CI/CD pipeline with blue/green deployment



Workshop

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Logistics

Instructions:

https://cicd-with-aws-fargate-lambda-bg.workshop.aws

Raise your hand if you have any questions. Feel free to work together with folks at your table.

High-five your neighbors and have fun! Please fill out feedback forms and follow the cleanup instructions once you are done.

We created an AWS account for you! Access it using your printed hash here:

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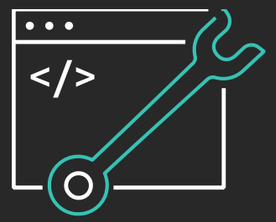

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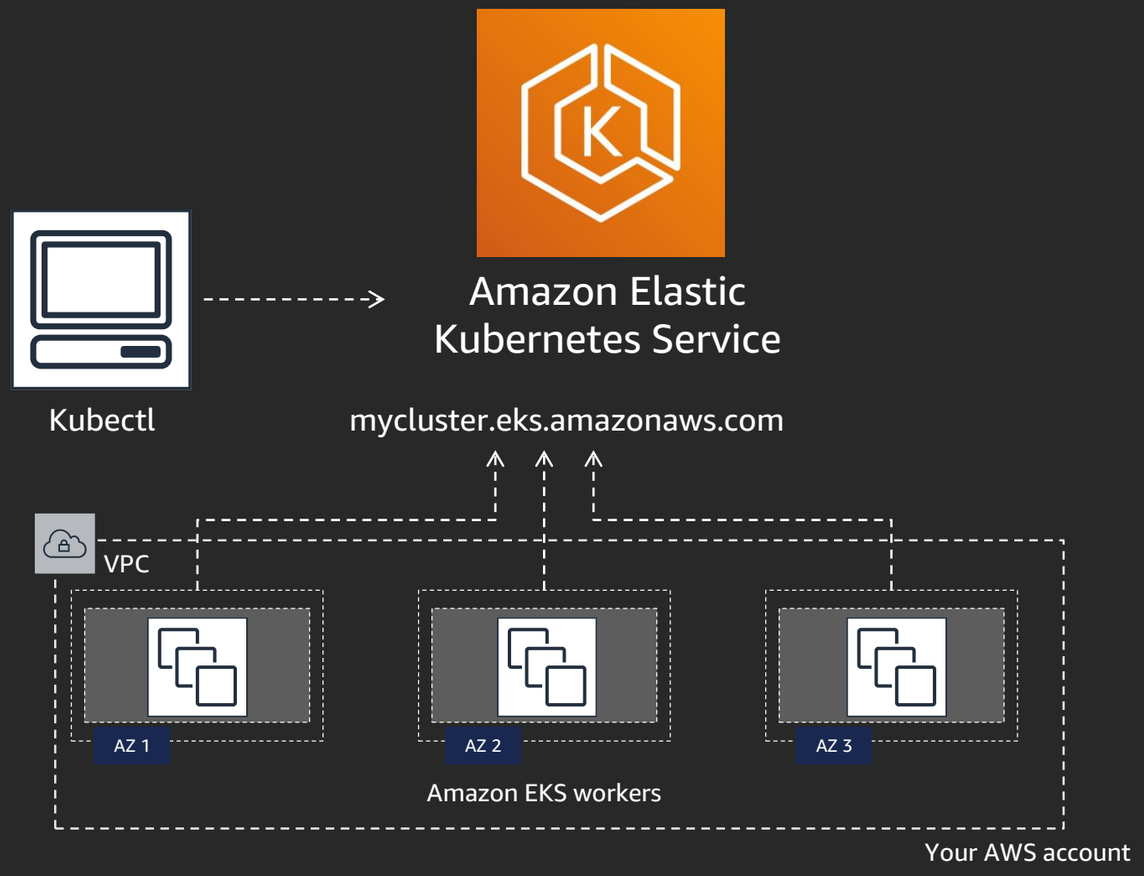
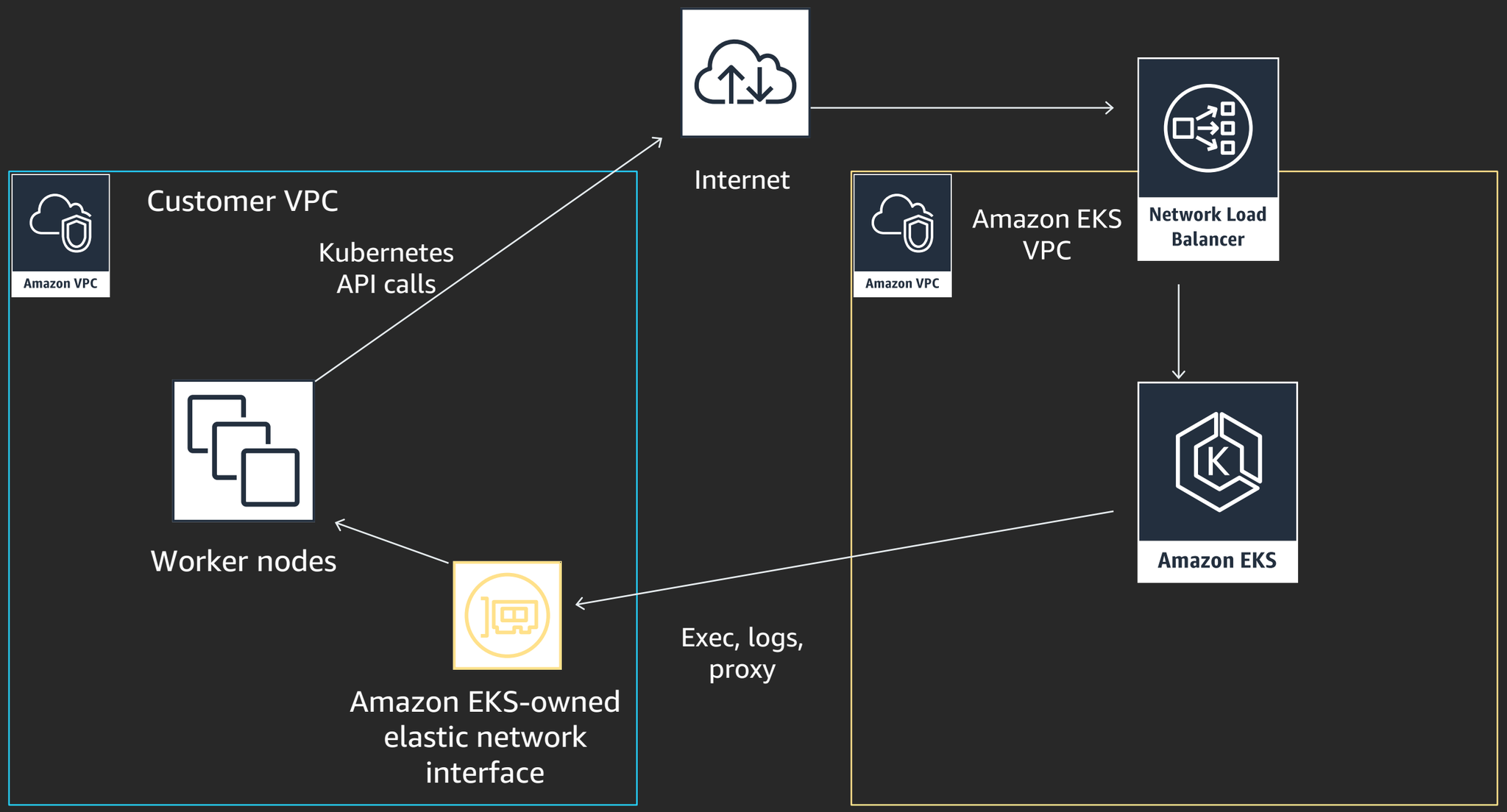
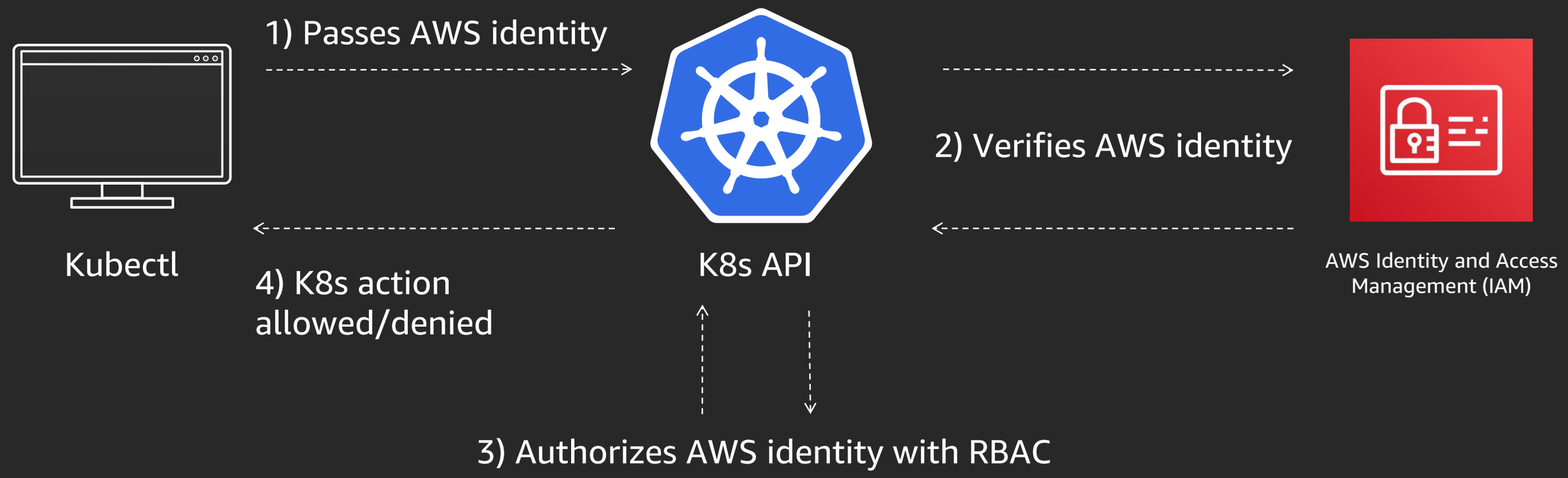
Introduction to Amazon Elastic Kubernetes Service

- No control plane to manage
- Secure by default
- Conformant and compatible
- Optimized for cost using Spot Instances
- Supports Windows and Linux containers
- Built with the community
- Amazon EKS supports Kubernetes versions 1.11, 1.12, 1.13, and 1.14
- Currently, no native Fargate support



Amazon EKS architecture



AWS Identity and Access Management (IAM) authentication